Does D-Flow help? What the data show

D-Flow can reach high Task 2 success, but the present results do not establish an advantage over tuned QGF. All eight D-Flow conditions have three frozen model-specific choices. Tuned original QGF and FK each scored 40/40 for all three models on fresh confirmation. These are selection/confirmation results, not a new heldout test or a causal comparison of averaging.

Own-best condition	lambda	Seed 11	Seed 23	Seed 37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	40/40
Stochastic / averaged Q	1	35/40	40/40	40/40
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	unqualified	unqualified	unqualified

The canonical F8 BC references scored 0/20 for each trained model; these cached controls have a separate role from the fresh40 selections above. Strong guided results do not isolate which mechanism caused improvement: cohorts, hyperparameters and selection differ. Equal numeric kappa across methods is not equal accumulated guidance. Episode starts are not additional trained seeds.

Your rule remains explicit: "Keep the all-33 requirement." Thirty of 33 guided choices are fixed; IPG3 remain unqualified. No new main100 heldout test or matched-kappa curve study has run. Cube has zero policy-performance episodes.

All 24 D-Flow selected tuples

Identity surrogate guidance, full-F8 endpoints, T=1. I is latent updates; K is critic neighbors. The literal3 column is the original three fresh40 outcomes, in frozen order; all 72 full alternative tuples are supplied in CSV.

Seed	Mode/Q	L	kappa	eta	I	sigma	K	CONF	Literal3
11	reg/raw	0	0.1	0.1	256	-	-	35/40	35/31/21
11	reg/raw	1	40	0.03	256	-	-	28/40	24/28/0
11	sto/raw	0	5	0.1	128	-	-	40/40	40/24/22
11	sto/raw	1	5	0.1	128	-	-	37/40	37/23/10
11	reg/avg	0	1	0.1	256	0.003	16	38/40	36/38/21
11	reg/avg	1	64	0.03	256	0.5	4	25/40	24/25/24
11	sto/avg	0	5	0.1	128	0.003	16	40/40	40/29/24
11	sto/avg	1	5	0.03	128	0.003	16	35/40	35/26/26
23	reg/raw	0	5	0.1	256	-	-	40/40	40/40/40
23	reg/raw	1	64	0.03	256	-	-	40/40	40/40/40
23	sto/raw	0	80	0.03	256	-	-	40/40	40/40/22
23	sto/raw	1	80	0.01	128	-	-	40/40	40/33/30
23	reg/avg	0	16	0.1	256	0.3	4	40/40	40/40/40
23	reg/avg	1	80	0.03	128	0.5	16	40/40	40/40/40
23	sto/avg	0	64	0.1	128	0.5	4	40/40	40/40/40
23	sto/avg	1	80	0.1	256	0.01	4	40/40	40/40/40
37	reg/raw	0	0.1	0.1	256	-	-	37/40	37/34/25
37	reg/raw	1	5	0.1	128	-	-	40/40	40/22/0
37	sto/raw	0	5	0.1	128	-	-	39/40	39/39/32
37	sto/raw	1	5	0.1	128	-	-	40/40	40/24/4
37	reg/avg	0	0.1	0.1	256	0.003	4	37/40	37/36/23
37	reg/avg	1	5	0.03	256	0.003	4	40/40	40/26/24
37	sto/avg	0	3	0.1	256	0.003	4	40/40	40/34/30
37	sto/avg	1	5	0.03	128	0.003	4	40/40	40/25/25

DF24 original aggregate freeze SHA 19d8af37. Full model/configuration IDs, all tuple values and evidence hashes are in DF24_SELECTED_TUPLES.csv and RESULTS.json. No hidden fourth candidate or retuning is introduced by this report.

Domain: 20 kappas from 0 to 80; eta .001/.003/.005/.01/.03/.1; I1/2/4/8/16/32/64/128/256; averaged-Q sigma .003/.01/.03/.06/.1/.2/.3/.5; K4/16. The adopted hierarchy is9,240 settings per model, not 73,440 Cartesian settings.

QGF, FK and every literal alternative

Selections use only their own frozen fresh40 confirmation outcomes. Their inference integration grids differ: QGF M can differ from trained F8; FK keeps the original actor F8 while using its particle integration M.

Method	Seed	kappa	M	N	noise	Selected
QGF	11	16	64	1	0	40/40
QGF	23	320	8	1	0	40/40
QGF	37	16	16	1	0	40/40
FK	11	80	64	32	0.02	40/40
FK	23	80	32	32	0.02	40/40
FK	37	80	16	32	0.02	40/40

All Task 2 literal fresh40 alternatives

Method	Seed	Order	kappa	M	N	noise	CONF	Chosen
QGF	11	0	20	64	1	0	38/40	
QGF	11	1	16	64	1	0	40/40	yes
QGF	11	2	16	32	1	0	39/40	
QGF	23	0	320	16	1	0	40/40	
QGF	23	1	40	32	1	0	40/40	
QGF	23	2	320	8	1	0	40/40	yes
QGF	37	0	16	32	1	0	39/40	
QGF	37	1	16	16	1	0	40/40	yes
QGF	37	2	12	32	1	0	40/40	
FK	11	0	80	64	32	0.02	40/40	yes
FK	11	1	8	8	16	0	30/40	
FK	11	2	5	64	16	0	37/40	
FK	23	0	80	32	32	0.02	40/40	yes
FK	23	1	5	32	16	0.02	30/40	
FK	23	2	8	16	16	0	29/40	
FK	37	0	80	16	32	0.02	40/40	yes
FK	37	1	5	8	32	0.02	40/40	
FK	37	2	5	64	16	0	39/40	

Complete QGF Task2+Task5 candidate tables contain 648 VAL20 cells: 27 kappa x 4 M x 3 models x 2 tasks. The original M8 panels are referenced, not charged again as new joint-scan episodes. FK has 2160 TRAIN-score rows (720/model); only its declared shortlists have native success outcomes.

Totals: episodes are not score rows

Recorded quantity	Count	Meaning
New native VAL/CONF episodes	30,360	Campaign ledger; not the sum of duplicated metric views
Evaluated active policy action rows	16,185,534	Physical policy controls; model-free controls tracked separately
New main heldout episodes	0	All33 prerequisite remains unmet
DF TRAIN-score rows	27,720	9,240/model; no native success interpretation
DF native candidate metric views	420	348 VAL20 requests +72 fresh40 alternatives
FK TRAIN-score rows / native metric views	2,160 /30	Full 720/model scores; 21 VAL20 + 9 CONF views
QGF full selection cells / literal CONF	648 /18	Both Task2 and Task5, all 3 models
Guided choices fixed / required	30 /33	DF24 +QGF3 +FK3; IPG3 unavailable

What "all outputs" includes in this update

The CSV directory includes every recorded DF TRAIN score, all 420 DF native candidate metric views, all 72 literal alternatives, all 648 QGF cells, all 18 QGF confirmation alternatives, the FK full TRAIN grid and all 30 FK native metric views. All33 statuses and source/model identities are explicit. The source episode CSV index records hashes and roles; it does not create new independent episode credit.

Missing measurements are labelled, not invented: an untested Cartesian DF cell is not a zero; FK TRAIN scores are not 720 native rollouts; IPG has no qualified native grid; Cube has no policy utility grid. Opaque traces/checkpoints/raw TRAIN are externally indexed and were not decoded. Historical full assets remain in the unchanged checkpoint archive.

Own-best selection, same-kappa curves and heldout generalization answer different questions. The proposed 540 same-kappa points target 100 episodes each via 20-episode increments, but those curves and their new heldout roles remain unrun.

Harder Cube: training is complete, utility pending

CubeTriple Task3 uses 46D observations and 25D H5 action chunks (five 5D controls), unaugmented local 1M TRAIN data, seed 11,128 x 128 networks, full F8 and T1. This is a limited-data adapted benchmark, not the paper 100M-data replication.

Actual stage	Saved result	Scope
Fresh BC50k	50,000 actor/Adam updates; parent91a8	No engineering-update credit
Goal-specific IQL50k	50,000 Q/V updates; fullparent2935	Actor fullstate/RNG frozen; Task3 reward
E1 flat25 callbacks	69 predicates PASS	Interface proof, not sampler or utility
Controlled H5 comparison	31 predicates PASS;40 external controls	Model-free controlled-reset diagnostic
Official reset R0	19 predicates PASS;4 resets/8 internal	0external; equal observations,9 broader state differences
E2a f32 actual	128 cells:120 PASS /8 latent FAIL	No sampler qualification; historical f64 suffix unlaunched
Standalone reference64	3941 eligible FD PASS;2000 AD checks PASS	Separate direct-action probe;59nonsmooth/1000inactive NA
Saved24 reader	8 failing cases retained;296 bad latent coordinates	0action coordinates over original threshold; no clipping-cause proof
I1 first actual attempt	NameError in reference helper during2nd trace	0 baseline/capture B20 calls; no numeric repair evidence

The eight E2a latent failures and 895 f32 FD diagnostic flags remain. The standalone64 comparison does not test a full-F8 latent-composed gradient or the failed sampler endpoint. I1 stopped before querying a sampler because a helper module global was not configured; its 937 earlier PASS predicates concern earlier gates, not the missing capture. A narrow binding proposal was pending at this frozen snapshot.

TRAIN rewards are per-cube goal count minus3:927,047 at-3;71,460 at-2;1,493 at-1; zero joint-goal rows. There are996,000 full-H5 eligible starts. Coverage is descriptive: neither proof of impossibility nor true IQL Q error. No hidden full-goal BC-success cutoff is imposed.

The H5 reward is sum(j=0..4) gamma^j r[j], with bootstrap gamma^5 M5 V(next), gamma=.999. Thus the nominal continuing-return range remains [-3000,0], rather than -15/(1-gamma). The last logged incoming-state TRAIN head0 Q mean was -1503.859, not a fresh final-checkpoint accuracy or Q-truth measurement.

Unresolved engineering and task evidence

Known-Q high-dimensional fixtures

All 596 engineering configurations ran across D8/32/128:585 PASS and11 original-QGF formula-parity FAIL. These are numerical checks on balanced analytic fixtures, not native returns or measured IQL critic error. The largest D32 case387 action difference was1.86511415 (625/640 coordinates), while its kappa0 control passed. Saved named exports do not prove hidden consumer values or a compiler/hardware cause. The large prospective comparison pilot is not adopted.

Original IPG

The materialized-reference B144 gate retains 134 PASS /10 FAIL. All 144 old production baselines reproduced 19 public fields (2736 byte proofs). Old/new failure sets overlap 7 cases, with 3 old-only and 3 new-only. Small residuals do not replace trajectory checks. Original 33 FAIL and 15 missing learned cases remain; 11 old failures occur at rho .1. Earlier dtype/lowering failures and both ridges are preserved. No IPG C/native selection is admitted.

Task5 QGF and D-Flow

Seed	Selected QGF kappa	M	Fresh CONF
11	96	16	40/40
23	2	64	36/40
37	80	32	40/40

Task5 tuned QGF scored 40/40, 36/40, 40/40. Four failures on seed 23 are real but do not establish a general collapse or averaging repair. Expanded Task5 DF still has four latent-only engineering failures; no expanded native DF utility comparison has run. Custom-reset exposure, zero standard withheld-action exposure and independently tuned task returns are distinct evidence.

The frozen all 33 status table includes every unavailable IPG slot. Neither the synthetic known-Q suite nor harder Cube changes that primary requirement. Failures, missing records and historical versions remain preserved.

What Q averaging can and cannot do

Qbar(s,a) = (1/K) sum_j Q(s, clip(a + sigma*epsilon_j))

One flow endpoint receives K nearby critic queries. K is not K flow trajectories or additional physical data. Sigma sets the per-coordinate neighborhood width; its Euclidean radius scales roughly as sigma*sqrt(D). The fixed antithetic bank is held throughout all I latent updates; K4 and K16 need not be nested.

A narrow false peak can be softened if nearby predictions are better. A broad wrong peak can survive. A legitimate narrow good peak can also be blurred. Finite K, clipping, min-head critics and the learned flow mean this is not exact Gaussian smoothing or an exact tilted/Boltzmann sampler. Identity guidance is a surrogate; lambda1 is latent prior pull, not an L1 norm or support detector.

Why performance remains pending on Cube

Actual BC and goal-specific IQL training have completed. The remaining blockers are concrete: qualify the original H5 sampler boundary and replay/reference contract, measure 20 original native environments co-resident with the full model during cold compilation, and then adopt the native H5/performance roles and measured budget. R0 two-environment RSS and E1 pure-F8 timing do not prove 20-env/4GB fit. A SOURCE-only BC boundary protocol was under review at this snapshot; it is not a performance run.

Official reset action spaces create fresh unowned RNGs. R0 found equal returned observation hashes but nine broader recognized-state differences. Same seed is not a proof of paired simulator starts. Any future performance protocol must disclose its reset law; it cannot silently reuse controlled pinned-Box pairing or reduce the physical cohort to fit.

What would establish a D-Flow benefit

Complete the required qualification and frozen selections, then use the prospectively declared heldout and matched-guidance comparisons with unchanged algorithms, roles and budgets. Current independently selected 40-episode panels are useful evidence, but do not establish a causal Qavg effect, true-Q repair, statistical dominance or heldout generalization.

Frozen scope, resources and evidence bundle

Plan365 quantity	Value
Known scientific outer-parent wall, once	45387.763027336 s
Separate old-admission UNKNOWN allowance	600 s; not measured actual wall
Used / initial 86,400 s tranche	45987.763027336 s
Remaining / active caps	40412.236972664 s /0
New main100 / Cube policy utility episodes	0 /0
All33 strict user answer	Keep the all-33 requirement

Administrative, child, watchdog and sampler clocks are not added to the scientific outer-parent clock. Model-free native controls and source-implied reset physics are separate ledgers, not policy VAL/CONF episodes. The initial 24 worker-hours are a reviewed tranche, not a whole-study completion guarantee. No automatic retry, scope reduction or heldout exception is created by this report.

Included and externally referenced

This compact update contains this PDF/TXT, machine-readable results, complete CSV exports, original CSV/JSON/source evidence copies and SHA manifests. The unchanged 35,845,587-byte private plan328 archive (2055 members; SHA 59b60fa9...) retains eight exact checkpoint files, six Task2 configurations, derived P1 data and the original 179 asset index. Raw 334,519,088-byte Cube TRAIN and the full historical 5.868 GB workspace archive remain external. This update is not a portable runtime or a whole-workspace transaction.

Frozen input: plan 365 SHA 7e8666c8... and I1 reconciliation 163f129c.... Later source reviews or executions belong to new dated receipts; this report will not be overwritten to incorporate them. No model, array payload, sampler, RNG or environment query was made to generate it. Every final PDF page must pass actual rendered visual QA, followed by root independent content and visual QA, before delivery.

Start with RESULTS.json, ALL33_GUIDED_SELECTION_STATUS_NO_HELDOUT.csv, the full candidate/grid CSVs and SCOPE_AND_MISSING_EVIDENCE_INDEX.csv. Source episode CSVs are indexed by exact path and hash, so duplicated metric views can be distinguished from fresh executions. All failures and unavailable cells remain explicit.